251020 1st Order Circuits

Voltage across a Capacitor:

$C = q/v$ [Capacitance ; F]

$i = dq/dt$ [Current ; A]

Combine these equations:

$$\Rightarrow i = dq/dt = C\, dv/dt \Rightarrow \boxed{i(t) = C\frac{dv(t)}{dt}}$$

Put in integral Form:

$$\Rightarrow \frac{dv}{dt} = \frac{1}{C} i \Rightarrow \int_{t_0}^{t} \left(\frac{dv}{dt'}\right) dt' = \frac{1}{C}\int_{t_0}^{t} i\, dt'$$

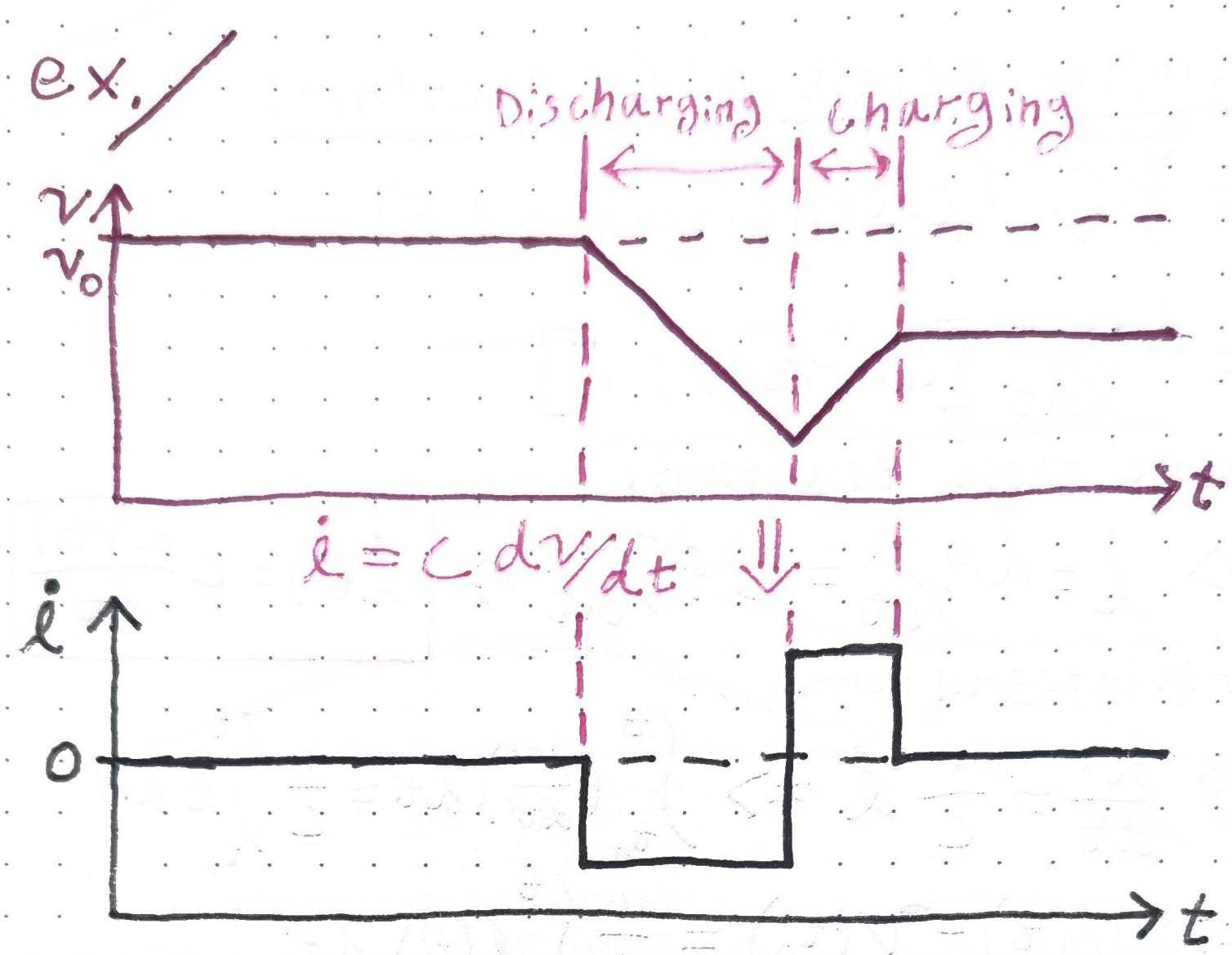
$$\Rightarrow V(t) - V(t_0) = \frac{1}{C}\int_{t_0}^{t} i(t')\, dt'$$

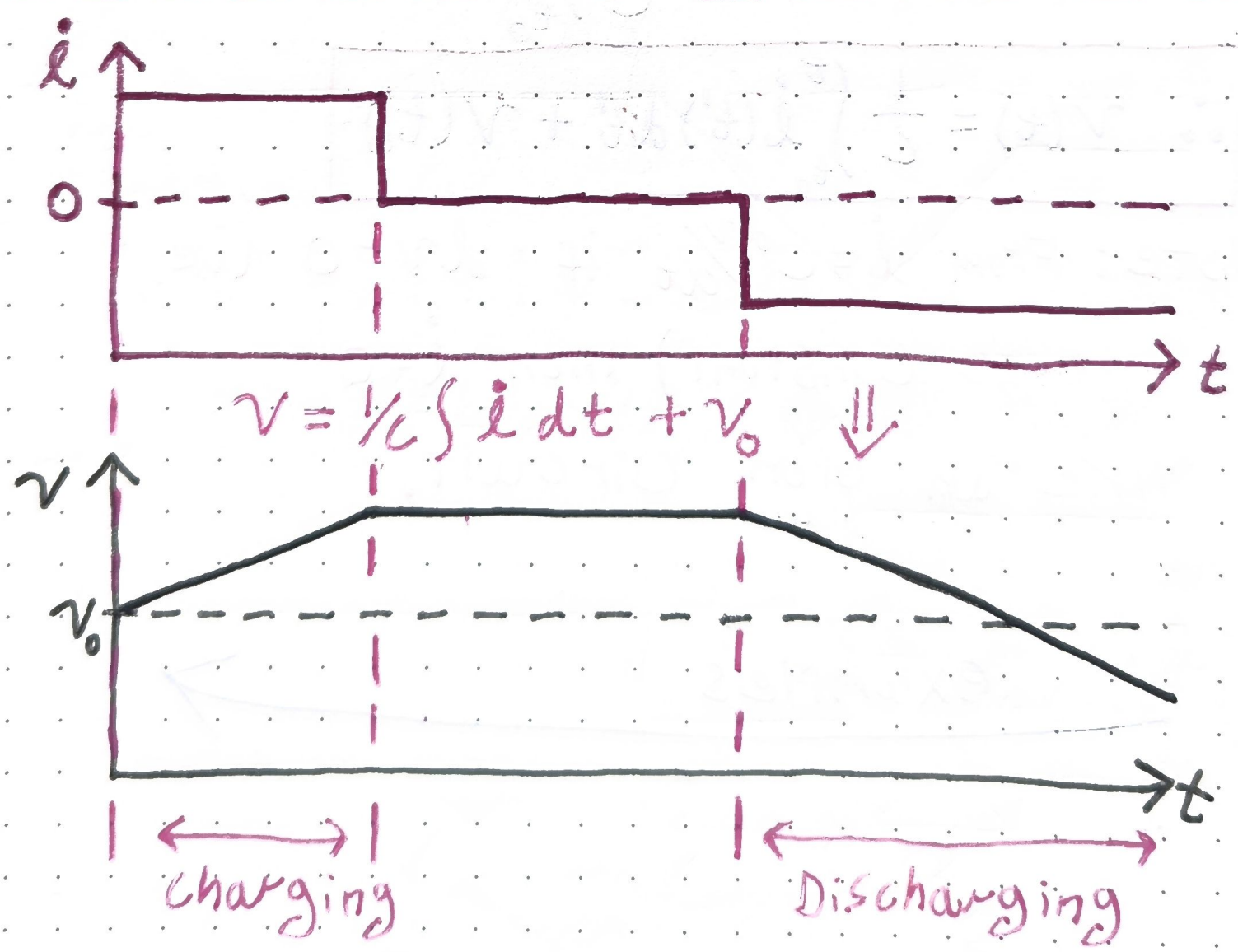
$$\boxed{\therefore\ v(t) = \frac{1}{C}\int_{t_0}^{t} i(t')\, dt' + v(t_0)}$$

Note: From $i = C\, dv/dt$ if $dv = 0$ (i.e., Voltage Constant) then $i = 0$, like an open Circuit.

examples →

ex./


Discharging
charging
v
v_0
t
$i = C\, dv/dt$
i
0
t


i
0
t
$v = 1/C \int i\, dt + v_0$
v
v_0
t
charging
Discharging

Current through an inductor:

$L = \Lambda / i$ [Inductance; H]

$v = \frac{d\Lambda}{dt}$ [Voltage; V]

Combine these equations:

$$v = \frac{d\Lambda}{dt} = L\frac{di}{dt} \Rightarrow \boxed{v(t) = L\frac{di(t)}{dt}}$$

Put in integral Form:

$$\Rightarrow \frac{di}{dt} = \frac{1}{L}v \Rightarrow \int_{t_0}^{t}\left(\frac{di}{dt'}\right)dt' = \frac{1}{L}\int_{t_0}^{t} v\, dt'$$

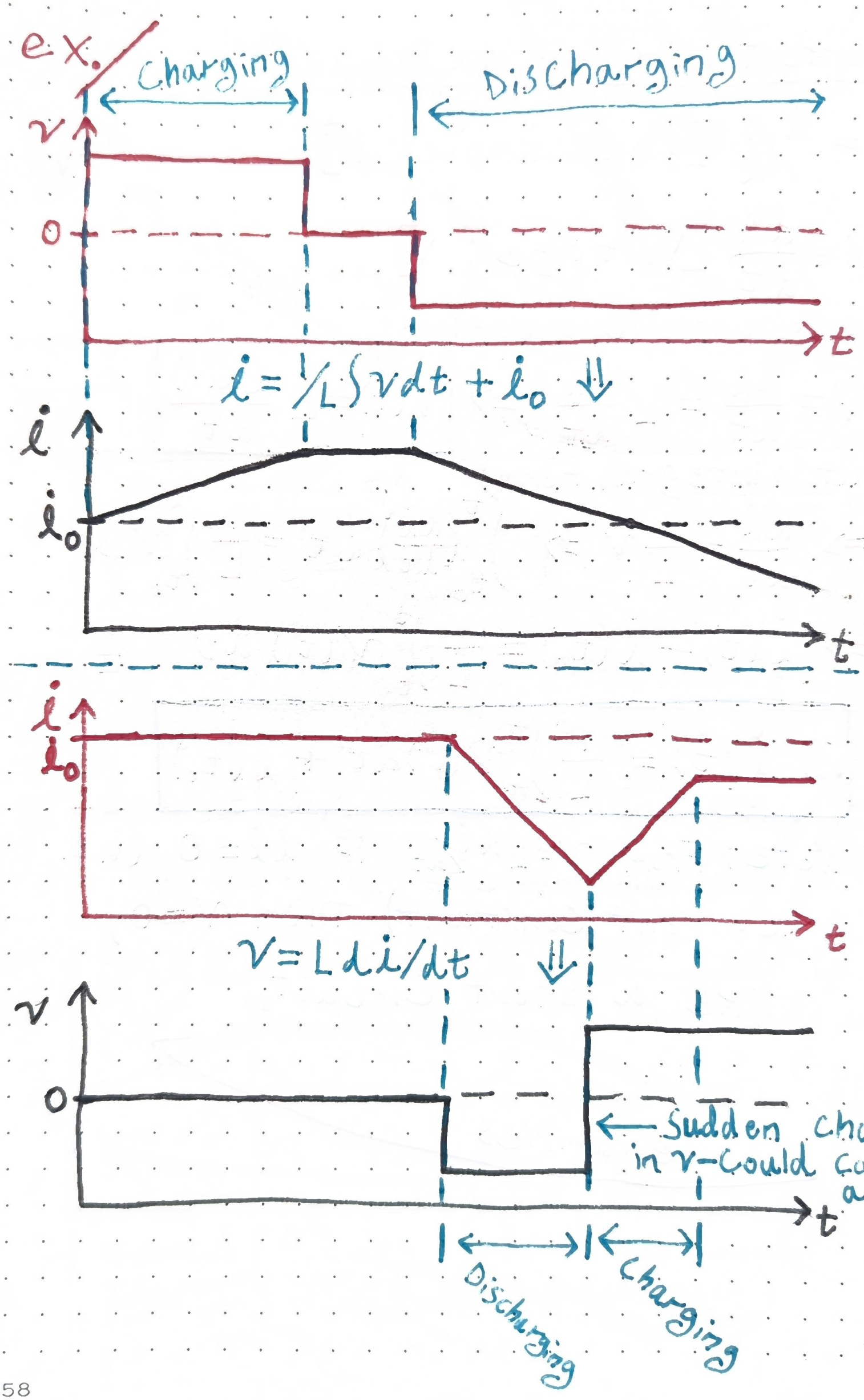
$$\Rightarrow i(t) - i(t_0) = \frac{1}{L}\int_{t_0}^{t} v(t')\, dt'$$

$$\boxed{\therefore\ i(t) = \frac{1}{L}\int_{t_0}^{t} v(t')\,dt' + i(t_0)}$$

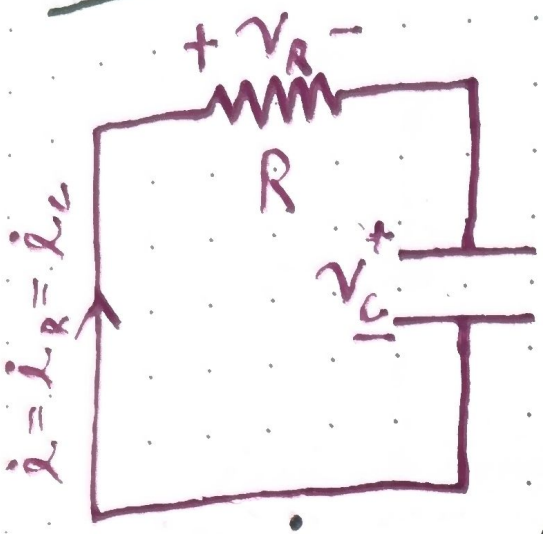
Note: From $v = L\, di/dt$ if $di = 0$ (i.e., current constant) then $v = 0$, like a short circuit.

examples

ex.


Charging
Discharging
v
0
t
$i = \frac{1}{L}\int v\,dt + i_0$ ⇓
i
i_0
t
i
i_0
t
$v = L\,di/dt$ ⇓
v
0
Sudden change in v - Could cause a spark
t
Discharging
Charging

1st Order RC Circuit:



Apply KVL:

$$v_R + v_c = 0$$

Apply Ohm's Law:

$$i_R R + v_c = 0$$

Apply $i_c = C\, dv_c/dt$ and $i_R = i_c$

$$\Rightarrow RC\frac{dv_c}{dt} + v_c = 0 \qquad \boxed{\therefore \frac{dv_c}{dt} + \frac{1}{RC}v_c = 0}$$

Initial Condition: $v_c(0^-) = V_s$
(Meaning the capacitor starts charged w/ voltage V_s)

Find Solution for v_c:

$$\Rightarrow \frac{dv_c}{dt} = \frac{-1}{RC}v_c \Rightarrow \int \frac{1}{v_c}\, dv_c = \frac{-1}{RC}\int dt$$

$$\Rightarrow \ln(v_c) = \frac{-1}{RC}t + a' \Rightarrow v_c = e^{-t/RC}e^{a'}$$

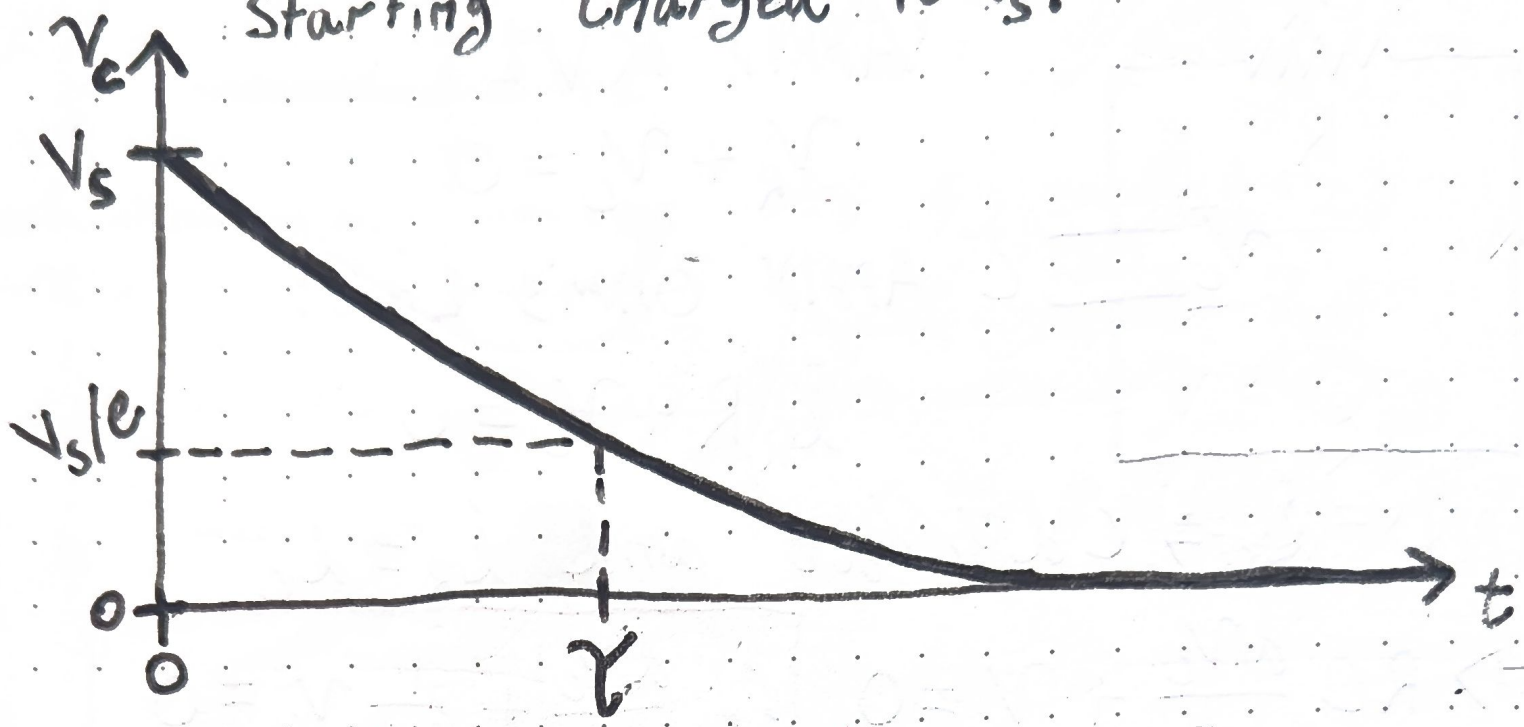
let $a = e^{a'}$ and $\boxed{\tau = RC \text{ [time-constant; s]}}$

$$\Rightarrow v_c = a e^{-t/\tau} \Rightarrow V_s = a e^{-0/\tau} \Rightarrow a = V_s$$

$$\boxed{\therefore v_c(t) = V_s e^{-t/\tau}}$$

Plot

Voltage across a capacitor (w/ C) discharging through resistance R after starting charged to V_s.



Note: $1/e \simeq 0.37$ (a litte more than a third)

$\Rightarrow$ Rule of thumb: After τ time has past, the capacitor is about 2/3 of the way to its final state (in terms of voltage)